

To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define BUFFER_SIZE 1024

int main() {
    char input_buffer[BUFFER_SIZE];

    printf("=====\n");
    printf(" Skill 2 - Interactive Shell Loop & Exit Handling\n");
    printf(" Type commands or 'exit' / 'quit' to terminate. \n");
    printf("=====\n\n");

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        if (fgets(input_buffer, sizeof(input_buffer), stdin) == NULL) {
            printf("\n[EOF detected] Exiting shell.\n");
            break;
        }

        input_buffer[strcspn(input_buffer, "\n")] = '\0';

        if (strlen(input_buffer) == 0) {
            continue;
        }

        if (strcmp(input_buffer, "exit") == 0 || strcmp(input_buffer, "quit") == 0) {
            printf("[Exit condition met] Exiting myshell cleanly. Goodbye!\n");
            break;
        }

        printf("[Processed Input]: \"%s\" (Length: %lu chars)\n", input_buffer,
            strlen(input_buffer));
    }

    return 0;
}
```

```
}
```

```
Last login: Fri Aug 14 14:25:57 on ttys000
[varnikakomali@VARNIKAs-MacBook-Air ~ % cd OS_LAB
[varnikakomali@VARNIKAs-MacBook-Air OS_LAB % cd Skill
[varnikakomali@VARNIKAs-MacBook-Air Skill % nano skill2.c
varnikakomali@VARNIKAs-MacBook-Air Skill % gcc -Wall skill2.c -o skill2
[./skill2
myshell> hello operating systems
You entered: hello operating systems
myshell> status_check
You entered: status_check
myshell> exit
Exiting shell...
varnikakomali@VARNIKAs-MacBook-Air Skill %
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>
```

```
#define MAX_BUFFER 1024
```

```
struct termios orig_termios;
```

```
void disableRawMode() {
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &orig_termios);
}
```

```
void enableRawMode() {
    tcgetattr(STDIN_FILENO, &orig_termios);
    atexit(disableRawMode);
    struct termios raw = orig_termios;
    raw.c_lflag &= ~(ECHO | ICANON);
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}
```

```
int main() {
    char buffer[MAX_BUFFER];
    int index = 0;
    char c;
```

```
    enableRawMode();
```

```
    printf("=====\n\r");
    printf(" Skill 2 (Part 2) - Raw Terminal & Buffer Input \n\r");
    printf(" Handles Backspace, Enter, and Multi-Char Cmds \n\r");
```

```

printf("  Type 'exit' to quit.                \n\r");
printf("=====\\n\\r\\n\\r");

printf("raw-shell> ");
fflush(stdout);

while (1) {
    if (read(STDIN_FILENO, &c, 1) <= 0) break;

    if (c == '\n' || c == '\r') {
        buffer[index] = '\0';
        printf("\\r\\n");

        if (strcmp(buffer, "exit") == 0) {
            printf("Exiting raw shell. Goodbye!\\r\\n");
            break;
        }

        if (index > 0) {
            printf("[Executed Command]: \"%s\" (Length: %d)\\r\\n", buffer, index);
        }

        index = 0;
        printf("raw-shell> ");
        fflush(stdout);
    } else if (c == 127 || c == '\b') {
        if (index > 0) {
            index--;
            printf("\\b \\b");
            fflush(stdout);
        }
    } else if (c >= 32 && c <= 126) {
        if (index < MAX_BUFFER - 1) {
            buffer[index++] = c;
            putchar(c);
            fflush(stdout);
        }
    }
}

return 0;
}

```

```
[varnikakomali@VARNIKAs-MacBook-Air Skill % nano skill2_p2.c
[varnikakomali@VARNIKAs-MacBook-Air Skill % gcc skill2_p2.c -o skill2_p2
[varnikakomali@VARNIKAs-MacBook-Air Skill % ./skill2_p2
=====
Skill 2 (Part 2) - Raw Terminal & Buffer Input
Handles Backspace, Enter, and Multi-Char Cmds
Type 'exit' to quit.
=====

[raw-shell> hellooooo
[Executed Command]: "hellooooo" (Length: 8)
[raw-shell> exit
Exiting raw shell. Goodbye!
varnikakomali@VARNIKAs-MacBook-Air Skill % █
```